Design and Implementation of a Modular Data Pipeline for GNN-Based Stock Diversification Analysis

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***Abstract*—This paper presents a modular data pipeline for constructing a unified dataset to support Graph Neural Network (GNN)-based stock diversification analysis. The pipeline integrates heterogeneous data sources, including financial market data, macroeconomic indicators, commodity prices, and relational company information. It follows a medallion architecture and focuses on the transformation and harmonization of data into a consistent, GNN-compatible schema, comprising entities, features, observations, and relations. The resulting dataset enables a graph-based representation of economic systems, combining fundamental and relational information beyond traditional price-based approaches.**

***Index Terms*—Graph Neural Networks, Data Pipeline, Financial Data, Portfolio Diversification**

I. Introduction

The objective of equity investments is to achieve high returns while minimizing risk. A common strategy to reduce portfolio risk is diversification, which essentially means to distribute capital across multiple assets. As shown by Markowitz, the effectiveness of diversification depends on the correlation structure among assets: highly correlated assets provide limited risk reduction, whereas low correlations can significantly reduce overall portfolio risk [1]. In practice, these correlations are typically estimated from historical returns. However, market dynamics and inter-firm relationships are more complex than what past performance alone can capture. This limitation motivates the incorporation of additional information sources and more fundamental modeling approaches. One way to address this is to view the economy as a network of interconnected actors that influence one another. Within this perspective, GNNs provide a natural framework for modeling economic entities and their relationships. By representing the economy as a graph, GNNs enable the learning and utilization of relational dependencies, thereby facilitating the identification of clustered risks within a portfolio. To effectively model such an economic structure, a comprehensive data foundation is required. This includes not only representations of relevant entities, such as companies, commodities, and states, but also data capturing the relationships between them, including ownership structures, product categories and other interdependencies. However, the necessary information is typically distributed across multiple heterogeneous data sources rather than being available from a single provider. Consequently, it is necessary to integrate and harmonize these data sources into a unified, GNN-compatible format. The objective of this work is therefore to design and implement a modular data pipeline that integrates financial and economic data from heterogeneous sources and transforms them into a unified database structure tailored for GNN-based diversification analysis.

II. Data Sources

A. *Yahoo Finance*

Yahoo Finance is a widely used financial data platform that provides access to a broad range of market and company-specific information. In this work, Yahoo Finance serves as primary data source, providing currency time series, market data such as stock prices, trading volumes, and shares outstanding, as well as company-level financial statements including balance sheets, income statements, ownership structures and dividend information.

B. *Stock Exchange Websites*

Since Yahoo Finance primarily relies on exchange-specific ticker symbols as identifiers for data retrieval, these tickers constitute the entry point to the data pipeline. However, no comprehensive global registry of listed companies and their ticker symbols exists. To address this limitation and ensure a geographically diverse and representative set of companies, ticker data is sourced directly from the official listings of major stock exchanges. For the initial construction of the pipeline, ticker symbols are collected from NASDAQ, New York Stock Exchange, and Euronext. The listing tables provide the ticker symbols required for querying Yahoo Finance. In addition, Euronext provides the International Securities Identification Number (ISIN).

C. *Global Macro Data*

To capture macroeconomic drivers at country level, data from the Global Macro Database is incorporated. This dataset aggregates information from 27 contemporary sources, including institutions such as the World Bank and the International

Monetary Fund (IMF), as well as 94 historical datasets. In total, it provides 46 macroeconomic variables at the national level, covering key dimensions such as gross domestic product, inflation, public finances, and labor market indicators. The temporal coverage spans from as early as 1086 up to forecast horizons extending to 2030, encompassing 240 countries worldwide [2].

D. International Monetary Fund

Since interest rates and commodity prices exert a substantial influence on economic activity, data from IMF is incorporated. In particular, the Primary Commodity Price System (PCPS) and the Monetary and Financial Statistics – Interest Rates (MFS_IR) datasets are used. The PCPS dataset provides detailed time series on global commodity prices, including energy resources (e.g., crude oil and natural gas), agricultural goods (e.g., wheat and coffee), and metals. The MFS_IR dataset contains a range of country-level interest rate indicators, including policy rates, lending rates, deposit rates, and interbank rates. These variables capture monetary conditions and reflect the cost of capital across different economies.

E. Wikidata

Wikidata is a large, collaboratively maintained knowledge graph that provides structured information on a wide range of entities and their relationships. In this work, it is primarily used to extract relationships between companies, including subsidiary links, ownership structures, product affiliations, group memberships, and industry classifications.

III. Data Pipeline Architecture

The data pipeline is structured according to a three-layer model, commonly referred to as the medallion architecture [3]. In this work, the pipeline is implemented up to the Silver layer, as illustrated in Fig. 1. The Bronze layer is responsible for extracting data from the respective sources and storing it in its raw form. Transformations at this stage are kept to a minimum and are only applied where necessary to ensure that the data can be stored in a suitable and consistent format within the database. The Silver layer is responsible for transforming and harmonizing the data, as well as structuring it into a GNN-compatible format. In addition, it enriches the data with metadata to facilitate categorization and improve usability, thereby enabling more informed decisions in the following Gold layer. The objective of the Silver layer is to establish a foundational database that supports a broad range of GNN architectures, while minimizing the introduction of biases that may limit the applicability of different modeling approaches to the underlying problem. The Gold layer, in contrast, focuses on tailoring the data to a specific modeling approach. At this stage, model-specific pre-processing steps are applied, including strategies for handling missing values, transforming the data into the required graph representation which means defining node and edge features, applying appropriate normalization techniques, and performing further transformations depending on the chosen GNN architecture. As the primary goal of this work is to establish a robust and flexible data foundation for subsequent experimentation with different GNN approaches, the pipeline is implemented only up to the Silver layer. The Gold layer is intentionally left for future work, where it can be adapted to the specific requirements of the selected modeling architecture without imposing premature assumptions. This layered architecture ensures a clear separation of concerns between data ingestion, general-purpose transformation, and model-specific processing, thereby improving modularity and maintainability. This is particularly well suited for this work, as it enables the construction of a flexible and extensible data foundation that can support a wide range of GNN-based approaches without imposing constraints on downstream modeling decisions.

A. Bronze Layer

The Bronze layer consists of one dedicated module for each data source from which data is extracted. To ensure consistency and maintain a clear separation of concerns, all modules follow a uniform structure. Each module comprises a Client class responsible for retrieving data from the respective source, a Transformer class that converts the data into a standardized long format, a Datahub class that manages all database-related operations, and an Ingestor class that orchestrates the overall ingestion process, including tasks such as batching where necessary.

1) Ticker Symbols from Stock Exchanges: To extract the ticker symbols, the Client module retrieves listings from the respective exchanges via CSV-based data endpoints. Since stock exchanges list a wide range of financial instruments, including derivatives, exchange-traded funds (ETFs), American Depositary Receipts (ADRs), and preferred shares, the transformer module filters the listings to retain only common stock tickers. This ensures that only entities providing a direct and consistent representation of firm-level financial performance are included. As ticker symbols are not globally unique identifiers, additional transformation steps are required. For securities listed on Euronext, exchange-specific suffixes are appended to ensure compatibility with Yahoo Finance conventions. Furthermore, Euronext provides ISINs, which are stored alongside ticker symbols as globally unique identifiers. This is particularly valuable for subsequent integration with Wikidata, where ISINs facilitate reliable entity resolution. In addition, corresponding Wikidata exchange identifiers are assigned to each ticker to further support entity matching in later steps. Finally, the harmonized ticker lists from all exchanges are consolidated and stored in a DuckDB table, serving as the foundational input for the Yahoo Finance module.

2) Financial Data from Yahoo Finance: Yahoo Finance provides two primary types of data in this pipeline: company-level financial data and currency exchange rate information. The Yahoo Finance Client utilizes the yfinance library to retrieve data. This library offers structured access to various data categories, including balance sheets, historical price data, dividends, and ownership information such as mutual fund holdings. The Ingestor module orchestrates the retrieval and

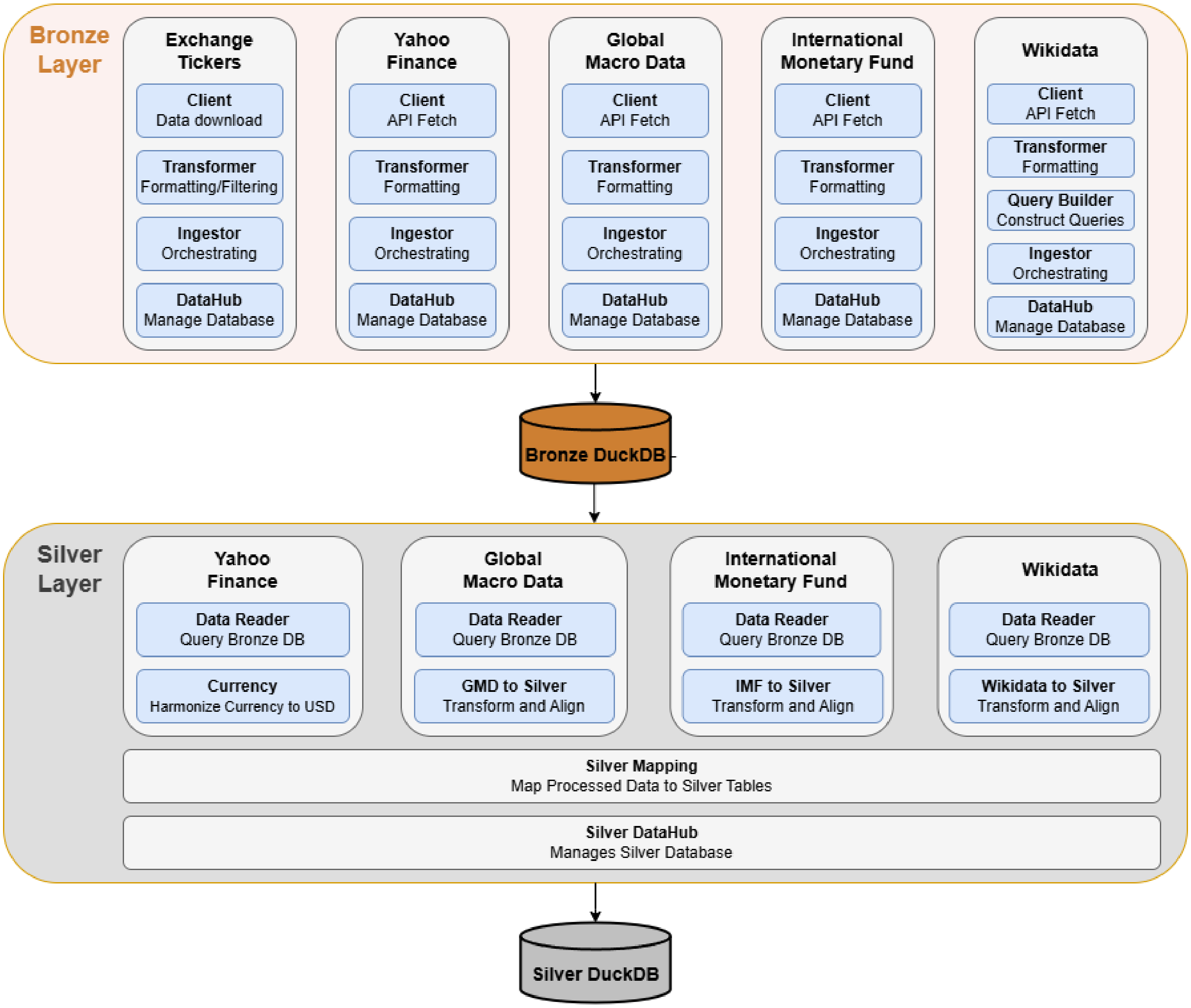

Fig. 1. Overview of the data pipeline architecture.

transformation process for each ticker. For company-level data, the retrieved information is passed to the Transformer module, where it is converted into a unified long format. This design choice is motivated by the high degree of heterogeneity in financial data across firms. Differences in data availability, temporal coverage, and reporting structures, which arise from varying industries and accounting standards, make a fixed schema impractical. The long format provides a flexible and consistent representation, enabling all observations to be stored using a standardized set of attributes, such as ticker, date, feature category, feature description, value, and unit. After processing company-level data, the Ingestor performs a second retrieval step to collect currency exchange rates based on the reporting and market currencies associated with the collected companies. This information is required for subsequent currency harmonization in the Silver layer as well as features within a graph. Finally, both company-level data and currency exchange rates are stored in separate tables within a DuckDB database in the Bronze layer.

3) Macroeconomic Data from Global Macro Database: The Bronze module for global macroeconomic data is comparatively simple, as the dataset is already well-structured and prepared. The Client module accesses the data via the global_macro_data Python library. Since the data is provided in a single tabular format, it can be directly transformed into a unified long format and stored in a dedicated DuckDB table.

4) Commodity and Interest Data from IMF: Since both interest rate and commodity price data are sourced from the International Monetary Fund, they are processed within a single module. The Client is implemented using the sdmx Python library, which provides access to the IMF's statistical data services, through which both datasets can be retrieved. To query commodity price data, specific indicator keys referencing individual commodities are required. These indicators are obtained from the IMF's code lists. Accordingly, the Ingestor first triggers the Client to retrieve all available commodity indicators, which are then transformed into an iterable structure by the Transformer. Based on this, the pipeline systematically queries price data for all available commodities at a global level with monthly frequency. To ensure comparability across different commodities, all prices are retrieved as index values rather than absolute price levels. In contrast, interest rate data does not require explicit indicator selection. Instead, a bulk query is performed to retrieve all available interest rate series for all countries at monthly intervals, including policy rates, lending rates, and deposit rates. All retrieved data is subsequently processed by the Transformer module and converted into a unified long format to store the data in two separated DuckDB tables.

5) Relational Company Data from Wikidata: Retrieving data from Wikidata requires substantially more effort compared to the other data sources. Unlike the remaining Client modules, which use reliable structured data endpoints or libraries, data access in Wikidata is performed via SPARQL queries. In Wikidata, each entity is identified by a unique identifier (QID). As the primary company data source in this pipeline is based on exchange-specific ticker symbols, an initial mapping step is required to resolve these tickers to their corresponding QIDs. This is achieved by querying Wikidata using ticker symbols in combination with exchange identifiers obtained from the ticker module, or ISINs if they are available. Due to the community-maintained nature of Wikidata, coverage is not guaranteed, and not all companies present in the Yahoo Finance dataset can be matched to corresponding entities. Once the QIDs are obtained, they are used to query the relevant relational information for each entity. To ensure stable query execution, data retrieval is performed in small batches of QIDs. For each type of information, batches of up to 25 QIDs are constructed. A dedicated query builder component is used to dynamically insert the QIDs into predefined SPARQL query templates before execution by the Client module. Given the variability and occasional instability of the Wikidata query service, batching is combined with a retry mechanism to handle failed requests and ensure robust data retrieval. Finally, the retrieved data is processed by the Transformer module, converting it into a unified long format for storage in a dedicated DuckDB table within the Bronze layer.

B. Silver Layer

The Silver layer loads the data from the respective tables in the Bronze DuckDB database and performs source-specific transformation and harmonization steps within modules dedicated to each data source and the respective data category they deliver. The primary objective is to integrate heterogeneous data sources and data types into a unified and consistent database schema tailored for subsequent use in different GNN architectures. The resulting schema is designed to represent both entities and their associated attributes as well as relations between entities in a structured manner. It comprises a central entity table that contains all possible entities in the dataset, including companies, countries, currencies, and products, which can be interpreted as nodes in the graph representation. In addition, a feature definition table provides an overview of all available features that can be associated with entities, such as financial statement variables, stock market indicators, and macroeconomic measures like GDP or interest rates. This table also includes metadata describing each feature, such as its category and the number of available observations. The actual feature values are stored in a separate observation table, which links entities to their corresponding features. While most features in the dataset are available as time series, no strict temporal harmonization across data sources is performed. Instead of enforcing a uniform frequency, the dataset is designed to preserve a reasonable high temporal resolution for each source. This design choice is motivated by the nature of diversification analysis, which focuses on comparing the current state of entities rather than modeling their temporal evolution or capturing dynamic correlations over time. By retaining the original frequencies, the dataset remains flexible for subsequent model construction, feature engineering, and the handling of missing values. This also enables the construction of more robust features, as it avoids the need to rely on noisy high-frequency time series data in cases where a stable snapshot representation of an entity is sufficient. To represent relationships between entities, the schema also includes a relation definition table, which captures different types of relationships along with associated metadata. The corresponding relation observations are stored in a dedicated table that links pairs of entities. All tables are interconnected through a set of relational keys, enabling the consistent linkage of entities, features, observations, and relations. This structure provides a flexible and scalable foundation for constructing graph-based representations of the data.

1) Yahoo Finance Data Transformation: Apart from type checking and format standardization, the most important harmonization step in the Silver module for Yahoo Finance data is currency normalization. As companies report financial data in different currencies, all monetary values are converted into a common reference currency, namely U.S. dollars (USD). Since each observation is associated with a specific reporting date, currency conversion can be performed using time-consistent exchange rates. This ensures that values are normalized consistently across both entities and time, enabling comparability within the dataset. In a second processing step, the data is prepared for integration into the Silver layer database schema. In this context, a distinction is made between entity-level features and relational information. While most features, such as financial statement variables or market indicators, are directly associated with individual entities, ownership information requires a different representation. Specifically, mutual fund holders and institutional holders are modeled as relations between entities rather than as entity features. This design choice reflects the heterogeneous and relational nature of ownership structures, where representing such information as simple attributes of a entity would be insufficient. The data is then stored accordingly to its assignment in the respective database table of the silver layer.

2) Global Macro Data Transformation: As the Global Macro Database is already provided in a standardized format, only limited preprocessing is required in the Silver layer. The main task is the selection of comparable and meaningful features. In this context, only features expressed in U.S. dollars or relative indicators (e.g., ratios based on GDP) are retained. Features reported in national currencies are excluded, as the additional complexity of currency conversion is not justified by their expected contribution to the analysis. Furthermore, forecast values included in the dataset are discarded, and the temporal scope is restricted to data from the year 2000 onwards. This decision is motivated by the focus on the current economic conditions relevant for diversification analysis. To improve interpretability and consistency, all selected features are enriched with unit information and assigned to predefined feature categories. As the dataset does not contain relational information suitable for graph edges, all data is represented as entity-level features. Consequently, the data is stored in the entity, feature definition, and observation tables of the Silver layer database.

3) IMF Data Transformation: For IMF commodity and interest rate data, several adjustments to the data representation are required. In particular, the names of commodities and interest rate series often include unit information, which is therefore extracted and stored in separate columns to ensure a consistent schema. Additionally, both commodity and interest rate series are categorized to improve interpretability and facilitate feature selection in later stages. For interest rate data, which is initially provided only with ISO3 country codes, country names are resolved by mapping to the Global Macro Database, which contains both identifiers. Commodity prices are typically provided in the form of index values which means relative to the base year 2016. As the objective is not to compare absolute price levels across different commodities but rather to analyze their temporal dynamics and their influence on companies, no further normalization is applied. In contrast, interest rate data is already consistently expressed as annual percentage rates and therefore does not require additional transformation. Since no relational information is present in this dataset, all variables are treated as entity-level features. Consequently, the data is stored in the entity, feature definition, and observation tables of the Silver layer database.

4) Wikidata Transformation: As Wikidata represents entities using QIDs, while the primary dataset relies on exchange-specific ticker symbols, an additional entity resolution step is required. In particular, relationships such as subsidiary links are returned as QIDs accompanied by entity labels. To ensure consistency with the rest of the dataset, these QIDs are mapped to corresponding ticker symbols wherever possible. Entities that cannot be matched to a ticker symbol are excluded from the dataset. This affects companies for which no corresponding financial data is available from Yahoo Finance, ensuring that all retained entities are properly represented. Furthermore, certain categories of relations are excluded if they provide only limited coverage, such as industry classifications, operating areas, or location-based relations. This filtering step improves the robustness of the resulting graph structure. As Wikidata is used exclusively to extract relational information, all retrieved data is stored in the relation definition and relation observation tables. In addition, newly identified entities, such as products, instances, or organizational groups, are incorporated into the entity table to extend the graph structure.

IV. Dataset Statistics and Overview

The final Silver layer of the data pipeline produces five database tables: one for unique entities, one for feature definitions, one for feature observations, one for relation definitions, and one for relation observations. The entity table contains 11,988 unique entities, which can be grouped into nine categories, as summarized in Table I. While some of these entities are associated with descriptive features that enrich their representation, others primarily serve a structural role within the graph.

TABLE I
Entities

Entity type	**Occurrence**	**Holds Features**
Companies	5776	True
Institutional holders	2971	False
Funds	1707	False
Products	864	False
Classes	279	False
States	193	True
Commodities	107	True
Groups	67	False
Currencies	24	True

In total, the features directly linked to entities comprise 7,562,734 observations. The majority of these observations originate from Yahoo Finance (7,253,602), followed by Global Macro Data (258,705) and the International Monetary Fund (50,427). Overall, the dataset includes 579 distinct features, which can be grouped into 36 subcategories, as shown in Table II. Since all the features are delivered in a time series manner they can have different frequencies and start dates.

The dataset contains 89,018 observations representing relations between entities. The majority of these relations are extracted from Yahoo Finance (80,952), while a smaller portion is derived from Wikidata (8,066). In total, the dataset comprises eight distinct relation types, as displayed in Table III

V. Discussion and Conclusion

This work presents a modular data pipeline designed to construct a foundational dataset for GNN-based diversification analysis. The pipeline integrates data from five heterogeneous sources to represent key economic entities and the relationships between them within a unified database schema. Its primary objective is to capture fundamental characteristics of relevant economic actors and their interconnections, which

TABLE II
ITEM CATEGORIES

Category	Observations	Entity	Frequency
balance_sheet	1,722,515	Companies	yearly
price_history	1,576,720	Companies	daily
cashflow	1,319,269	Companies	yearly
income_statement	1,202,174	Companies	yearly
shares_outstanding	1,080,953	Companies	daily
dividends	201,234	Companies	not defined
Currency_to_USD	128,085	Currencies	daily
government_finance	86,235	States	yearly
national_accounts	57,490	States	yearly
external_sector	45,992	States	yearly
major_holders	22,652	Companies	no timeseries
monetary_policy	17,247	States	yearly
prices	17,247	States	yearly
crisis	17,247	States	yearly
deposit_rates	16,089	States	yearly
lending_rates	12,702	States	yearly
policy_rates	6,148	States	yearly
classification	5,749	States	yearly
population	5,749	States	yearly
labor_market	5,749	States	yearly
government_rates	4,851	States	yearly
market_rates	3,728	States	yearly
commodity_index	1,764	Commodities	monthly
metals	945	Commodities	monthly
softs	819	Commodities	monthly
energy	693	Commodities	monthly
oils_and_oilseeds	567	Commodities	monthly
grains	378	Commodities	monthly
livestock	378	Commodities	monthly
precious_metals	252	Commodities	monthly
forestry	252	Commodities	monthly
food	189	Commodities	monthly
fertilizers	189	Commodities	monthly
other_commodities	189	Commodities	monthly
seafood	189	Commodities	monthly
other_rates	105	States	yearly

TABLE III
RELATION CATEGORIES

Relation	Observations	Source Entity	Target Entity	Edge Features
institutional_holders	42,291	Companies	Institutional Holders	Ownership (%)
mutualfund_holders	38,661	Companies	Funds	Ownership (%)
instance_of	5,465	Companies	Class	No
products	1,422	Companies	Products	No
part_of	874	Companies	Groups	No
owned_by	210	Companies	Companies	No
subsidiaries	58	Companies	Companies	No
investments	37	Companies	Companies	No

influence the performance of publicly traded companies. This structured representation enables the application of various GNN models. A key strength of the resulting dataset lies in its focus on incorporating multiple economically relevant factors beyond purely price-based information. In contrast to traditional approaches that rely heavily on historical price data, this dataset emphasizes fundamental indicators that describe the economic performance and structural characteristics of companies. This allows for a more comprehensive and informative representation of the underlying drivers of diversification. In addition, the harmonization of units within entity-level features ensures comparability across entities and data sources, which is essential for meaningful analysis. The overall data structure further contributes to the flexibility of the approach. The clear separation between entities, observations, and relations, combined with a flexible database schema, enables the construction of graph representations that can be adapted to different GNN architectures. In particular, distinguishing between entity related and relational information allows for a GNN-suitable modeling of complex economic systems. This is complemented by the modular design of the pipeline, which makes it easy to integrate new data sources and add further stock exchanges to retrieve data from a broader range of companies across the world. Despite these strengths, several limitations of the current dataset must be acknowledged. The number of available relations is relatively small and concentrated on ownership structures, which ignores dependencies from supply chains and other connections. Furthermore, Wikidata, which serves as the primary source for additional relational data, is inherently unstable and varies in quality due to its community-driven nature. Its limited coverage further reduces the number of usable relations. Another limitation concerns the coverage of the dataset, which is currently restricted to a small number of stock exchanges located in the United States and Europe. This limits the global representativeness of the data introduces a geographical bias of the companies and currencies available in the dataset. Moreover, the feature space is largely composed of numerical variables, with only limited inclusion of categorical features, potentially restricting the ability of certain GNN architectures to capture richer semantic information. Future work should therefore focus on extending both the breadth and depth of the dataset. A key direction is the inclusion of additional stock exchanges to improve global coverage and increase the diversity of represented entities. Expanding the set of relational data is equally important, particularly by incorporating more stable and higher-quality data sources beyond Wikidata in order to enrich the graph structure. Furthermore, greater emphasis should be placed on integrating categorical features to complement numerical data and provide additional context for graph-based models. Finally, the implementation of the Gold layer represents an important next step, enabling the transformation of the dataset into task-specific graph representations tailored to choosen GNN architecture to analyze diversification of stock portfolios.

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